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GATE CSE · DATABASE MANAGEMENT SYSTEM · CHAPTER-WISE PYQS

File Structures and Indexing

GATE CSE Database Management System · Previous Year Questions with Answer Key

29	24	1998 – 2026	English
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Marks 1	15 questions
Marks 2	14 questions
Answer key	at the end

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Marks 1 15 questions

Q1. GATE CSE 2026 Set 2

Open solution

In the context of DBMS, consider the two sets $\mathbf{T}$ and $\mathbf{S}$ given below.

T	S
I: Logical schema	L: Views
II: Physical schema	M: File organization and indexes
III: External schema	N: Relations

Which one of the following is the correct match from T to S ?

(A) $\mathrm{I}-\mathrm{L}$, II -M , III -N

(B) $I-M, II-L, III-N$

(C) $\mathrm{I}-\mathrm{N}, \mathrm{II}-\mathrm{M}$, III -L

(D) $\mathrm{I}-\mathrm{N}, \mathrm{II}-\mathrm{L}, \mathrm{III}-\mathrm{M}$

Q2. GATE CSE 2024 Set 2

Open solution

Which of the following file organizations is/are I/O efficient for the scan operation in DBMS?

(A) Sorted

(B) Heap

(C) Unclustered tree index

(D) Unclustered hash index

Q3. GATE CSE 2024 Set 1

Open solution

In a B+ tree, the requirement of at least half-full (50%) node occupancy is relaxed for which one of the following cases?

(A) Only the root node

(B) All leaf nodes

(C) All internal nodes

(D) Only the leftmost leaf node

Q4. GATE CSE 2021 Set 2

Open solution

A data file consisting of 1,50,000 student-records is stored on a hard disk with block size of 4096 bytes. The data file is sorted on the primary key RollNo. The size of a record pointer for this disk is 7 bytes. Each student-record has a candidate key attribute called ANum of size 12 bytes. Suppose an index file with records consisting of two fields, ANum value and the record pointer to the corresponding student record, is built and stored on the same disk. Assume that the records of data file and index file are not split across disk blocks. The number of blocks in the index file is _____

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Q5. GATE CSE 2021 Set 1

Open solution

Consider a linear list based implementation in a file system. Each directory is a list of nodes, where each node contains the file name along with the file metadata, such as the list of pointers to the data blocks. Consider a given directory foo.

Which of the following operations will necessarily require a full scan of foo for successful completion?

(A) Renaming of an existing file in foo
(B) Opening of an existing file in foo
(C) Deletion of an existing file from foo
(D) Creation of a new file in foo

Q6. GATE CSE 2019

Open solution

Which one of the following statements is NOT correct about the B^+ tree data structure used for creating an index of a relational database table?

(A) B^+ tree is a height-balanced tree
(B) Non-leaf nodes have pointers to data records
(C) Each leaf node has a pointer to the next leaf node
(D) Key values in each node are kept in sorted order

Q7. GATE CSE 2016 Set 2

Open solution

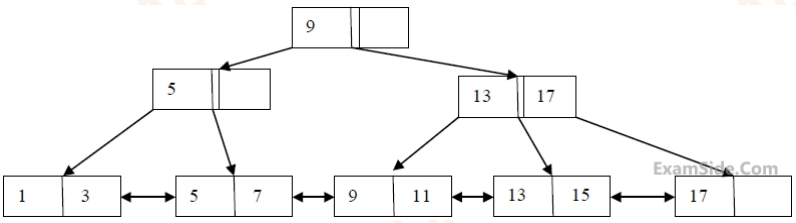
B^+ Trees are considered BALANCED because

(A) the lengths of the paths from the root to all leaf nodes are all equal.
(B) the lengths of the paths from the root to all leaf nodes differ from each other by at most 1.
(C) the number of children of any two non-leaf sibling nodes differ by at most 1.
(D) the number of records in any two leaf nodes differ by at most 1.

Q8. GATE CSE 2015 Set 2

Open solution

With reference to the B^+ tree index of order 1 shown below, the minimum number of nodes (including the Root node) that must be fetched in order to satisfy the following query: "Get all records with a search key greater than or equal to 7 and less than 15" is _____



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Q9. GATE CSE 2015 Set 1

A file is organized so that the ordering of data records is the same as or close to the ordering of data entries in some index. Then that index is called

(A) Dense (B) Sparse (C) Clustered (D) Unclustered

Open solution

Q10. GATE CSE 2010

Consider a B^+- tree in which the maximum number of keys in a node is 5. What is the minimum number of keys in any non-root node?

(A) 1 (B) 2 (C) 3 (D) 4

Open solution

Q11. GATE CSE 2008

A clustering index is defined on the fields which are of type

(A) non-key and ordering
(B) non-key and non-ordering
(C) key and ordering
(D) key and non-ordering

Open solution

Q12. GATE CSE 2005

A B-Tree used as an index for a large database table has four levels including the root node. If a new key is inserted in this index, then the maximum number of nodes that could be newly created in the process are:

(A) 5 (B) 4 (C) 3 (D) 2

Open solution

Q13. GATE CSE 2000

B^+-trees are preferred to binary trees in databases because

(A) Disk capacities are greater than memory capacities
(B) Disk access is much slower than memory access
(C) Disk data transfer rates are much less than memory data transfer rates
(D) Disks are more reliable than memory

Open solution

Q14. GATE CSE 1999

Which of the following is correct?

(A) B-trees are for storing data on disk and B^+ trees are for main memory.
(B) Range queries are faster on B^+ trees.
(C) B-trees are for primary indexes and B^+ trees are for secondary indexes.
(D) The height of a B^+ tree is independent of the number of records.

Open solution

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Q15. GATE CSE 1998

Open solution

There are five records in a database.

Name	Age	Occupation	Category
Rama	27	CON	A
Abdul	22	ENG	A
Jennifer	28	DOC	B
Maya	32	SER	D
Dev	24	MUS	C

There is an index file associated with this and it contains the values 1, 3, 2, 5 and 4. Which one of the fields is the index built from?

(A) Age (B) Name (C) Occupation (D) Category

Marks 2 14 questions

Q16. GATE CSE 2026 Set 2

Open solution

An index in a DBMS is said to be dense if an index entry appears for every searchkey value in the indexed file. Otherwise it is called a sparse index. Consider the following two statements.

S1: A hash index must be a dense index

S2: A B^{+}tree index can be a sparse index

Which one of the following options is correct?

(A) Both S 1 and S 2 are true (B) Both S 1 and S 2 are false

(C) S 1 is true and S 2 is false (D) S1 is false and S2 is true

Q17. GATE CSE 2025 Set 2

Open solution

In a B^{+}- tree where each node can hold at most four key values, a root to leaf path consists of the following nodes:

$A = (49, 77, 83, -), B = (7, 19, 33, 44), C = (20^{*}, 22^{*}, 25^{*}, 26^{*})$

The *-marked keys signify that these are data entries in a leaf.

Assume that a pointer between keys k_1 and k_2 points to a subtree containing keys in [k_1, k_2], and that when a leaf is created, the smallest key in it is copied up into its parent. A record with key value 23 is inserted into the B^{+}- tree.

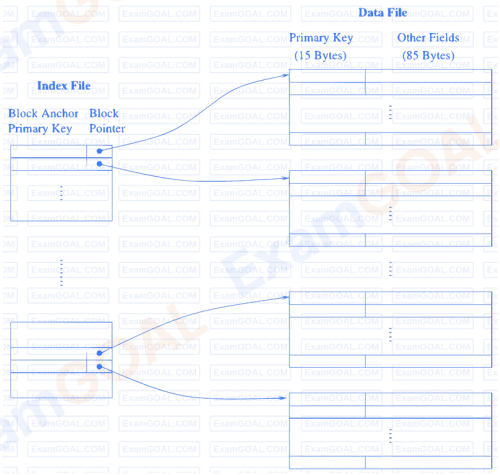
The smallest key value in the parent of the leaf that contains 25^{*} is _____ . (Answer in integer)

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Q18. GATE CSE 2023

Open solution

Consider a database of fixed-length records, stored as an ordered file. The database has 25,000 records, with each record being 100 bytes, of which the primary key occupies 15 bytes. The data file is block-aligned in that each data record is fully contained within a block. The database is indexed by a primary index file, which is also stored as a block-aligned ordered file. The figure below depicts this indexing scheme.



Suppose the block size of the file system is 1024 bytes, and a pointer to a block occupies 5 bytes. The system uses binary search on the index file to search for a record with a given key. You may assume that a binary search on an index file of b blocks takes $\lceil \log_2 b \rceil$ block accesses in the worst case.

Given a key, the number of block accesses required to identify the block in the data file that may contain a record with the key, in the worst case, is _____.

Q19. GATE CSE 2022

Open solution

Consider two file systems A and B, that use contiguous allocation and linked allocation, respectively. A file of size 100 blocks is already stored in A and also in B. Now, consider inserting a new block in the middle of the file (between 50th and 51st block), whose data is already available in the memory. Assume that there are enough free blocks at the end of the file and that the file control blocks are already in memory. Let the number of disk accesses required to insert a block in the middle of the file in A and B are n_A and n_B, respectively, then the value of $n_A + n_B$ is _____.

Q20. GATE CSE 2020

Open solution

Consider a database implemented using B+ tree for file indexing and installed on a disk drive with block size of 4 KB. The size of search key is 12 bytes and the size of tree/disk pointer is 8 bytes. Assume that the database has one million records. Also assume that no node of the B+ tree and no records are present initially in main memory. Consider that each record fits into one disk block. The minimum number of disk accesses required to retrieve any record in the database is _____.

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Q21. GATE CSE 2015 Set 3

Consider a B^+ tree in which the search key is 12 bytes long, block size is 1024 bytes, record pointer is 10 bytes long and block pointer is 8 bytes long. The maximum number of keys that can be accommodated in each non-leaf node of the tree is ____.

Open solution

Q22. GATE CSE 2009

The following key values are inserted into a B^+-tree in which order of the internal nodes is 3, and that of the leaf nodes is 2, in the sequence given below. The order of internal nodes is the maximum number of tree pointers in each node, and the order of leaf nodes is the maximum number of data items that can be stored in it. The B^+- tree is initially empty.

10, 3, 6, 8, 4, 2, 1

The maximum number of times leaf nodes would get split up as a result of these insertions is

(A) 2 (B) 3 (C) 4 (D) 5

Open solution

Q23. GATE CSE 2008

Consider a file of 16384 records. Each record is 32 bytes long and its key field is of size 6 bytes. The file is ordered on a non-key field, and the file organization is unspanned. The file is stored in a file system with block size 1024 bytes, and the size of a block pointer is 10 bytes. If the secondary index is built on the key field of the file, and a multi-level index scheme is used to store the secondary index, the number of first-level and second-level blocks in the multi-level index are respectively

(A) 8 and 0 (B) 128 and 6 (C) 256 and 4 (D) 512 and 5

Open solution

Q24. GATE CSE 2007

The order of a leaf node in a B^+- tree is the maximum number of (value, data record pointer) pairs it can hold. Given that the block size is 1K bytes, data record pointer is 7 bytes long, the value field is 9 bytes long and a block pointer is 6 bytes long, what is the order of the leaf node?

(A) 63 (B) 64 (C) 67 (D) 68

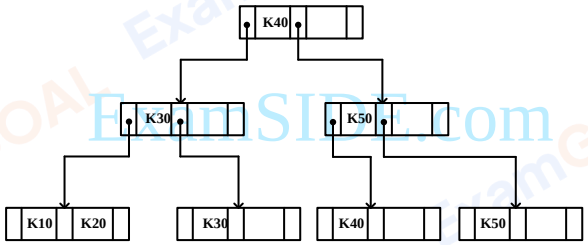
Open solution

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Q25. GATE CSE 2007

Open solution

Consider the B^+ tree in the adjoining figure, where each node has at most two keys and three links.



Keys K 15 and then K 25 are inserted into this tree in that order. Exactly how many of the following nodes (disregarding the links) will be present in the tree after the two insertions?



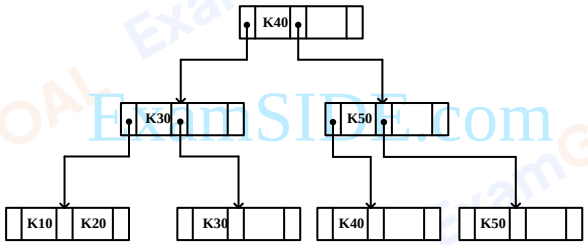
(A) 1 (B) 2 (C) 3 (D) 4

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Q26. GATE CSE 2007

Open solution

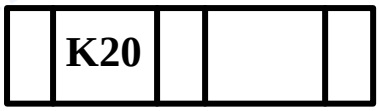
Consider the B^+ tree in the adjoining figure, where each node has at most two keys and three links.



Keys K15 and then K25 are inserted into this tree in that order. Now the key K50 is deleted from the B^+ tree resulting after these two insertions. Consider the following statements about the B^+ tree resulting after this deletion.

i) The height of the tree remains the same.

ii) The node



(disregarding the links) is present in the tree.

iii) The root node remains unchanged (disregarding the links).
Which one of the following options is true ?

(A) Statements (i) and (ii) are true
(B) Statements (ii) and (iii) are true
(C) Statements (iii) and (i) are true
(D) All the statements are false

Q27. GATE CSE 2004

Open solution

The order of an internal node in a B^+ tree index is the maximum number of children it can have. Suppose that a child pointer takes 6 bytes, the search field value takes 14 bytes, and the block size is 512 bytes. What is the order of the internal node?

(A) 24 (B) 25 (C) 26 (D) 27

Q28. GATE CSE 2004

Open solution

Consider a table T in a relational database with a key field K. A B-tree of order p is used as an access structure on K, where p denotes the maximum number of tree pointers in a B-tree index node. Assume that K is 10 bytes long; disk block size is 512 bytes; each data pointer P_D is 8 bytes long and each block pointer P_B is 5 bytes long. In order for each B-tree node to fit in a single disk block, the maximum value of p is

(A) 20 (B) 22 (C) 23 (D) 32

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Q29. GATE CSE 2002

A B^+ - tree index is to be built on the Name attribute of the relation STUDENT. Assume that all student names are of length 8 bytes, disk blocks are of size 512 bytes, and index pointers are of size 4 bytes. Given this scenario, what would be the best choice of the degree (i.e. the number of pointers per node) of the B^+ - tree?

Open solution

(A) 16 (B) 42 (C) 43 (D) 44

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Answer Key

Marks 1

1. **C**	2. **A, B**	3. **A**	4. **698**	5. **A, D**
6. **B**	7. **A**	8. **5**	9. **C**	10. **B**
11. **A**	12. **A**	13. **B**	14. **B**	15. **C**

Marks 2

16. **A**	17. **33**	18. **6**	19. **153**	20. **4**
21. **50**	22. **B**	23. **C**	24. **A**	25. **A**
26. **A**	27. **C**	28. **C**	29. **C**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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